

BILLAL HOSSAIN SHISHIR

Artificial Intelligence | Data Engineering | Software Development

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PROFESSIONAL SUMMARY

Final-year Bachelor of Information and Communication Technology student at the University of Tasmania, majoring in Artificial Intelligence, with hands-on experience across AI, data engineering, backend development, security, testing and project delivery. Built five live recruiter-facing portfolio systems backed by full engineering repositories using FastAPI, PostgreSQL, Docker, pytest and GitHub Actions. Project Manager for RAMIFY OS, coordinating requirements, testing, documentation, client feedback and final delivery.

CORE TECHNICAL SKILLS

Programming & Web: Python, C, JavaScript, SQL, HTML, CSS, Chart.js

AI & Data: pandas, NumPy, scikit-learn, TF-IDF, Logistic Regression, cosine similarity, Isolation Forest, anomaly detection, skill extraction

Backend & Databases: FastAPI, REST APIs, SQLAlchemy, Pydantic, PostgreSQL, SQLite, MQTT

Cloud, DevOps & Testing: Docker, Docker Compose, GitHub Actions, GitHub Pages, pytest, Git/GitHub, Nectar Cloud, AWS fundamentals, Linux, Windows

Security & Delivery: SHA-256, Ed25519, requirements analysis, functional testing, documentation, stakeholder communication, project coordination

SELECTED LIVE PROJECTS

AI Helpdesk Copilot | AI / Full-Stack / IT Support

[Live demo](#) | [GitHub](#)

- Built a full FastAPI, SQLAlchemy and PostgreSQL-ready helpdesk with a TF-IDF + Logistic Regression classifier across 7 support categories, transparent priority rules and retrieval over 70 troubleshooting articles.
- Implemented ticket creation, technician notes, similar-incident search, resolution analytics, 25 automated tests and an instant GitHub Pages recruiter demo backed by the complete repository implementation.

Real-Time Data Engineering Pipeline | Data Engineering / IoT

[Live demo](#) | [GitHub](#)

- Built a smart-building pipeline for 10 simulated devices with MQTT ingestion, Pydantic validation, transformation/enrichment, PostgreSQL persistence and dead-letter handling.
- Added rule-based, rolling 3-sigma and Isolation Forest anomaly detection, operational observability, Docker Compose infrastructure, automated tests and a live browser demo.

Network Monitoring & Incident System | IT Operations / Infrastructure

[Live demo](#) | [GitHub](#)

- Built a FastAPI monitoring platform for 5 simulated services with latency tracking, a configurable 3-consecutive-failure threshold, automatic incident creation and automatic recovery resolution.
- Added incident timelines, operational metrics, PostgreSQL/Docker support, automated tests and optional integration with the AI Helpdesk API.

SELECTED LIVE PROJECTS - CONTINUED

Document Verification & Trust System | Security / Cryptography

[Live demo](#) | [GitHub](#)

- Built a privacy-aware FastAPI verification platform using SHA-256 fingerprints without storing original document contents, with deterministic exact-copy verification and audit logging.
- Implemented Ed25519-signed JSON/PDF receipts, QR-based public verification, PostgreSQL/Docker support, automated tests and a recruiter-facing GitHub Pages demo.

Australian Job Intelligence Dashboard | Data Analytics / NLP

[Live demo](#) | [GitHub](#)

- Built a technology-job analytics platform using a curated 240-record Australian dataset, 32 tracked technical skills and 10 locations, with interactive market filters and Chart.js visualisations.
- Implemented transparent CV-to-role Technical Skill Coverage using deterministic skill normalisation, plus a FastAPI/PostgreSQL backend, Docker setup, tests and GitHub Pages demo.

PROFESSIONAL EXPERIENCE

Project Manager - RAMIFY OS | University Industry Project | Hobart, TAS Jul 2026 - Present

- Lead project planning and coordination for an auditable AI product-trust demonstration, aligning requirements, team responsibilities, testing, documentation and client feedback.
- Coordinate verification-focused workflows, architecture material, user guidance, scripted demonstrations and final presentation preparation across multiple trust scenarios.
- Work directly with the industry client to translate feedback into testable changes and delivery priorities.

Customer Service Representative | Bennett's Petroleum | Hobart, TAS Oct 2024 - Present

- Provide responsive customer service, handle transactions and resolve enquiries accurately in a fast-paced environment.
- Complete daily operational tasks independently while identifying recurring issues and contributing practical solutions.

Crew Member | McDonald's Rosny Park | Hobart, TAS Jun 2024 - Sep 2024

- Processed customer orders and transactions accurately and coordinated with colleagues during busy service periods.

Customer Service Assistant | Gadget and Gear | Dhaka, Bangladesh Apr 2023 - Dec 2023

- Supported customers with technology product enquiries, recommendations, payments, returns, complaint resolution and stock tasks.

INDUSTRY ENGAGEMENT

Open Innovation Hackathon | TasNetworks x University of Tasmania x AWS | Hobart, TAS May 2026

- Collaborated over three days to design and present a prototype for a real Tasmanian energy-sector challenge using rapid problem-solving and AI/cloud concepts.

DataEngBytes Melbourne 2026 | Industry Conference | Melbourne, VIC 23-24 Jul 2026

- Attended sessions on AI engineering, GraphRAG, Snowflake, synthetic data, observability, data architecture and agent workflows.

EDUCATION

Bachelor of Information and Communication Technology (BICT) - Major in Artificial Intelligence | University of Tasmania | Hobart, TAS Expected Nov 2026

International First Year Diploma Science | University of Tasmania | Hobart, TAS

International First Year Diploma Science | University of Tasmania | Hobart, TAS 2024

ADDITIONAL

Registration: Registration to Work with Vulnerable People (Tasmania) | **Visa:** No employer sponsorship required; full-time work rights upon course completion | **References:** Available on request